LITHO Mainnet 9005 Client Handoff

Last reviewed: 2026-07-29

Handoff status: **conditional acceptance requested**. The base chain and public node endpoints are live. The operational controls listed under "Open gates" must be assigned and closed; this document does not represent unconditional production acceptance.

1. Delivered service

Item	Delivered state
Network	LITHO mainnet
Cosmos/CometBFT chain ID	`lithosphere_9005-1`
EVM chain ID	`9005` (`0x232d`)
Native asset	`LITHO`; base denomination `ulitho`; 18 decimals
Fixed supply	`1,000,000,000 LITHO`
Genesis time	`2026-07-27T17:00:00Z`
Genesis SHA-256	`13e4875b4a9dddc63bdfbd4968c7265f9bbc49218b59c5b49231a56fa313046f`
Mainnet binary SHA-256	`0546677a9cf3a7f458797b65181a46f21c89185933e832d89ce728a144fd258c`
Height-1 block hash	`7418C1962B64597EE91D6747ECE3D5325C8B17B261E4C0E4A109A9BAFE74F509`
Initial validator set	One bonded validator, `validator1`, bonded with `1 LITHO`
Launch result	Launched and independently reverified on 2026-07-28

The fixed-supply binary enforces the exact genesis supply, prevents inflation from being re-enabled, and rejects completed transactions that would exceed the native supply cap. This is a consensus-critical project patch that passed focused internal tests; it has not received an independent third-party audit.

2. Client integration endpoints

Interface	Production endpoint	Expected identity
EVM JSON-RPC	`https://rpc-mainnet.litho.ai`	`eth_chainId = 0x232d`
EVM WebSocket	`wss://rpc-mainnet.litho.ai/websocket`	EVM chain `9005`
REST/LCD	`https://api-mainnet.litho.ai`	`lithosphere_9005-1`
gRPC over TLS	`grpc-mainnet.litho.ai:9090`	`lithosphere_9005-1`

Interface	Production endpoint	Expected identity
Read-only CometBFT	`https://rpc-mainnet.litho.ai/status`	`lithosphere_9005-1`
Final genesis	`https://rpc-mainnet.litho.ai/genesis.json`	SHA-256 above

POST `/` on the RPC hostname is EVM JSON-RPC. Only allowlisted, read-only CometBFT query routes are public. Broadcast and administrative CometBFT routes are not exposed through the HTTPS proxy.

RPC and REST terminate on sentry 1. gRPC terminates on sentry 2 because port `9090` on sentry 1 is occupied by an existing service. The certificates installed at handoff expire on 2026-10-26; Certbot timers and Nginx reload hooks were validated at activation.

Handoff-day public recheck

At `2026-07-29T06:44Z`, two sequential public status queries advanced from height `174683` to `174711`, reported `catching_up=false`, and identified `lithosphere_9005-1`. EVM JSON-RPC returned `0x232d`; REST returned `lithosphere_9005-1`; height 1 matched the sealed block hash; and the downloaded genesis body and its response header both matched the approved SHA-256.

At handoff time, the same unauthenticated check did not accept Lithoscan as publicly ready because its root request returned HTTP 403. That historical gate was closed on 2026-07-31: the production cutover, public smoke tests, block progression monitoring, clock-skew re-verification, and rollback closeout all passed without requiring rollback.

3. Deployed topology

Role	Host	Service	Mainnet home	Exposure
Validator	`194.5.157.233`	`lithod-mainnet-9005-val`	`/var/lib/litho-mainnet-9005-val`	Node APIs loopback-only; P2P through private mesh
Sentry/RPC 1	`31.97.39.146`	`lithod-mainnet-9005-sentry`	`/var/lib/litho-mainnet-9005-sentry`	Public node and proxy traffic
Sentry/RPC 2	`72.60.177.106`	`lithod-mainnet-9005-sentry`	`/var/lib/litho-mainnet-9005-sentry`	Public node and gRPC traffic

The nodes use the dedicated `wg-mainnet` mesh (`10.200.5.0/24`, UDP `51825`). This mainnet is VPS-only and has no AWS, KMS, or Secrets Manager dependency. Existing Makalu and Kamet services on the shared sentries are separate and must not be modified during mainnet work.

4. Evidence and delivered artifacts

Artifact	Purpose
Launch record	Sealed identity, launch steps, live verification, and endpoint activation
Final genesis and checksum	Canonical network bootstrap artifact
Fixed-supply review	Internal review of the consensus-critical supply patch
Validator backup status	Offline identity-backup evidence and checksum

Artifact	Purpose
Restart incident record	Impact, root cause, recovery, and follow-up
Topology and deployment plan	Host, port, isolation, and execution details
Operations runbook	Current mainnet checks and recovery guardrails
Ownership and admin-control handover	Accountable ownership, transferred controls, Safe applicability, and validator-key replacement boundary
Acceptance checklist	Evidence review, owner assignment, and sign-off
Lithoscan cutover record and closeout report	Certificate installation, deployment identity, public cutover, smoke tests, synchronization, and rollback decision

The tracked handoff package does not contain SSH private keys, validator private keys, WireGuard private keys, recovery keys, certificates' private keys, passwords, or API tokens. Transfer those only through the client's approved secret manager or an authenticated out-of-band channel.

5. Scope boundary

Included in this handoff:

- sealed and checksum-pinned genesis;
- three-node VPS topology with a private validator/sentry mesh;
- one active bonded validator;
- fixed-supply mainnet binary and deployment automation;
- TLS node endpoints and read-only public CometBFT routes;
- production Lithoscan mainnet explorer and restricted rollback controls;
- two client-confirmed offline encrypted copies of the initial validator identity backup; and
- launch and incident evidence.

Not enabled or not included:

- Bridge, Swap, Faucet, and MultX;
- additional bonded validators or an active validator failover host;
- a remote signer or HSM; and
- an independent third-party audit of the fixed-supply patch.

6. Open gates

Priority	Gate	Required closure
Critical	One validator is the entire active validator set	Client accepts the availability risk and approves a multi-validator or controlled failover roadmap. Never run a second signer with the same consensus key.

Priority	Gate	Required closure
Critical	Recurring live signing-state backup is not enabled (`backup_enabled: false`)	Implement encrypted backups of the current `priv_validator_state.json`, test restoration without starting a signer, define retention, and name two custodians. The launch-time height-0 identity package is not a current-state backup.
High	Mainnet-specific Prometheus targets and no-block alert are not present in the repository monitoring configuration	Add all three nodes, page when no block is committed for more than two minutes, test routing, and assign 24/7 response ownership.
High	Raw sentry node ports remain available during integration	Restrict raw RPC/REST/gRPC/EVM ports after consumers move to TLS endpoints; retain required P2P access.
High	Restore, failover, and transaction drills remain unrecorded	Execute controlled drills and retain evidence before claiming full operational readiness.
High	Consensus-critical fixed-supply patch has internal review only	Obtain an external review or record explicit client risk acceptance.
Medium	Allocation wallet purposes and remaining governance/economic parameters are not fully recorded	Supply auditable labels, fee/burn decisions, and governance owners.
Medium	EVM chain ID registry acceptance record is outstanding	Complete and archive the external registry submission/acceptance evidence.

7. Handoff decision

The client may accept the running base-chain service while tracking the open gates as post-handoff obligations, or withhold production acceptance until the Critical and High items are closed. The decision, named owners, target dates, and any explicit risk acceptances must be recorded in CLIENT_ACCEPTANCE_CHECKLIST.pdf.